TEAM ONBOARDING & OVERVIEW

MOL Bubi Expansion & Infrastructure Tool

A data-driven decision system recommending optimal bike-lane corridors and new station sites across Budapest.

CORE OBJECTIVE
Actionable Urban Recommendations

SCOPE
Bike Lanes & Station Growth

DELIVERABLE
Interactive Decision Dashboard

PROJECT VISION

What Are We Building & Why?

1. Priority Bike Corridors

Detecting high-use road segments currently lacking dedicated cycle lanes.

2. Optimal Station Sites

Identifying underserved population hubs beyond current station coverage.

3. Scenario Comparison

Evaluating network impact across existing vs. proposed expansion models.

URBAN CONTEXT

The Core Challenges We Are Solving

1. **Disconnected Network Gaps**
   Key riding routes lack continuous bike lanes, forcing riders into heavy mixed traffic.

2. **Coverage Limits Outside Center**
   Outer residential areas lack nearby Bubi stations within comfortable walking distance.

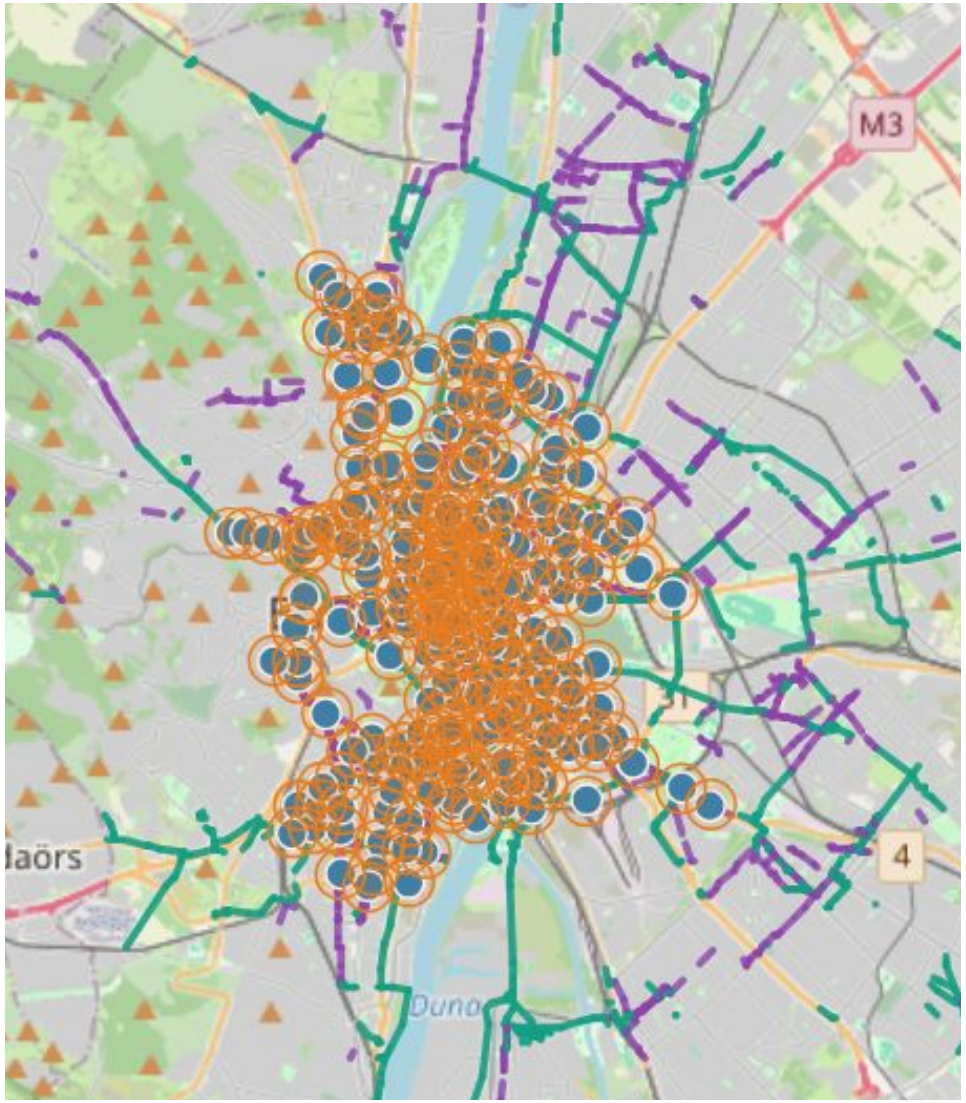
3. **Complex Planning Choices**
   City planners need clear visual evidence before committing capital infrastructure.

Target Impact

Transparent, data-backed evidence for city stakeholders and planners.

DATA LAYER 1

Existing Stations & Cycling Network



Active Docking Stations

220 Stations

Separate Cycleways

1,923 Segments

On-Road Bike Lanes

2,595 Segments

** Baseline uses a 400m walking catchment radius to measure current station reach.*

DATA LAYER 2

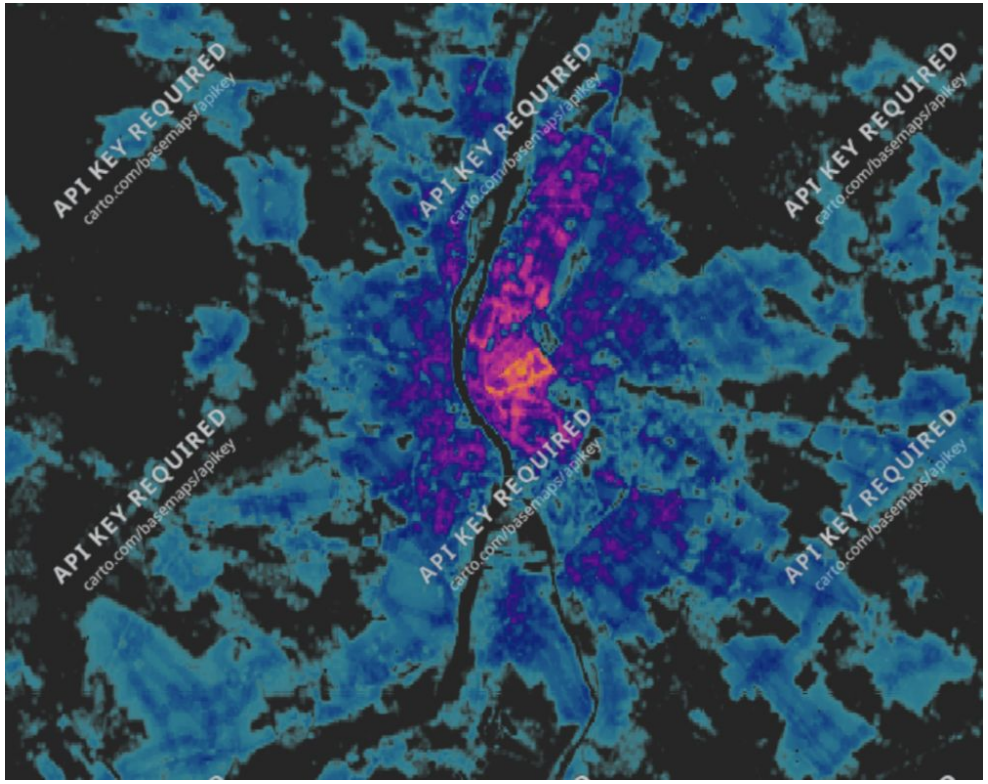
Urban Population Density Heatmap

Mapping Unserved Demand

We integrate gridded population estimates (WorldPop / GHSL) to pinpoint high-density residential zones.

Smart Spatial Grid

Divided into H3 hexagon cells to measure exact density outside current station catchment areas.



SYSTEM ARCHITECTURE

How the Recommendation Pipeline Works

STEP 01

Data Ingestion

Imports OpenStreetMap street graphs, Bubi stations, and population grids.

STEP 02

Slope-Aware Routing

Simulates travel trips across pairs, preferring flat/downhill routes over steep hills.

STEP 03

Gap Detection

Identifies high-volume road directions that currently lack dedicated bike lanes.

STEP 04

Scoring & Ranking

Ranks candidate corridors and top station locations for clear visual presentation.

MODULE 1 OVERVIEW

Finding High-Priority Bike Lane Corridors

Per-Direction Assessment

Evaluates road directions independently to ensure lanes are recommended where cyclists actually ride.

Slope & Terrain Awareness

Routing accounts for elevation so uphill climbs are naturally avoided in simulated trips.

Corridor Continuity

Merges adjacent missing segments into continuous, logical infrastructure projects.

CORRIDOR SCORING LOGIC

Route Volume × Gap Length × Continuity

Prioritizes road sections carrying heavy cycling traffic with continuous missing links.

MODULE 2 OVERVIEW

Selecting Optimal New Station Sites

FILTER RULE

Outside Walking Reach

Candidates must lie beyond the 300–400m walking network threshold of existing stations.

STRATEGY A

Lane-Aware Siting

Combines population density with direct connectivity to existing bike paths.

STRATEGY B

Population-Only Siting

Focuses purely on maximizing human density regardless of current bike lane presence.

SCENARIO ENGINE

Comparing Network Scenarios

SCENARIO 1

Baseline Network

Uses only existing Bubi stations and current cycling routes.

Benchmarking Point

SCENARIO 2

Lane-Aware Growth

Adds proposed stations connected to existing bike paths.

Integrated Network

SCENARIO 3

Density Expansion

Adds proposed stations in highest density population hubs.

Maximum Catchment

FINAL DELIVERABLE

Interactive Visual Dashboard

Dynamic Map Layers
Toggle routes, population heatmaps, bike lanes, and station coverage.

Live Scenario Switcher
Compare baseline vs. lane-aware vs. population expansion seamlessly.

Data Export Capabilities
Download top corridor rankings and station coordinates in standard formats.

Built for Stakeholders

Designed to present clear, transparent rationale to university deans, professors, and city transportation planners.

Streamlit | PyDeck / Folium

IMMEDIATE FOCUS

Team Roadmap & Next Steps

WEEKS 1–2

Data Ingestion & OSM

Set up dataset snapshots, population grids, and standardized tables.

WEEKS 3–4

Routing & Scoring Engine

Build slope-aware trip router, gap detector, and station scoring logic.

WEEKS 5–7

Dashboard & Demo

Assemble scenario toggle UI, export layers, and final presentation.

Let's build a clean, impactful decision-support system for Budapest!